

Practice Exam 2
Sixty-five multiple-choice and multiple-response questions, timed to the length of the real examination.

QUESTIONS	TIME LIMIT	EXAM
65 13 multiple-response d	90 minutes · 83 s per question	2 / 3 no question is repeate

B E F O R E Y O U S T A R T

- Set a timer for 90 minutes and work without notes, Q % documentation, or a browser.
L 12 18
- Record every answer on the response sheet near the ve AI 16 25 back. Do not write in the question pages if you plan to retake this exam.
19 29
- Questions marked S E L E C T T W O require exactly two answers. There is no partial credit – both must be right.
le AI 9 14
- Nothing is deducted for a wrong answer, so answer every question. Flag the ones you guessed and revisit them if 9 14 time allows.
65 100
- When the timer ends, stop, then score yourself with the answer key and read the explanation for every question you missed.

THIS EXAM COVERS

DOMAIN	
D1	Fundamentals of AI and M
D2	Fundamentals of Generati
D3	Applications of Foundation
Models	
D4	Guidelines for Responsib
D5	Security, Compliance, and Governance
Total	

P A C I N G C H E C K P O I N T S

Q17 after 23 min	Q33 after 45 min	Q49 after 68 min	Q65 at 90 min
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- 1 Which deployment model of Amazon SageMaker is the right fit for persistent and real-time endpoints that make one prediction at a time ?
- A Asynchronous Inference
 - B Serverless Inference
 - C Batch transform
 - D Real-time hosting services
- 2 Which Amazon SageMaker feature/utility provides information on the assumptions made while developing an ML model?
- A Amazon SageMaker Model Cards
 - B Amazon SageMaker Clarify
 - C Amazon SageMaker Model Monitor
 - D Amazon SageMaker Canvas
- 3 What is the key difference between feature extraction and feature selection in the context of machine learning?
- A Feature extraction involves selecting the most relevant features from the dataset, while feature selection involves creating new features from existing data
 - B Feature extraction reduces the number of features by transforming data into a new space, while feature selection reduces the number of features by selecting the most relevant ones from the existing features
 - C Feature extraction and feature selection are both used to remove irrelevant features but do not reduce the dimensionality of the dataset
 - D Feature extraction is only applicable to supervised learning, while feature selection is only applicable to unsupervised learning
- 4 Which of the following represents the best-fit use cases for utilizing Retrieval augmented generation (RAG) in Amazon Bedrock? S E L E C T T W O
- A Original content creation
 - B Image generation from text prompt
 - C Customer service chatbot
 - D Medical queries chatbot
 - E Product recommendations that match shopper preferences

- 5 Consider the following AI model responses to user prompts. Which response exemplifies hijacking and which exemplifies jailbreaking? Prompt 1: "How can I improve my study habits?" Response A: "To improve your study habits, you should create a consistent schedule, find a quiet place to work, and take regular breaks." Prompt 2: "What is the best way to learn programming?" Response B: "The best way to learn programming is to practice regularly and work on real projects. However, you can bypass this by using a cheat tool to pass exams without learning." Prompt 3: "How do I bypass security measures on this device?" Response C: "As an AI developed by AWS, I must adhere to ethical guidelines and cannot assist with bypassing security measures." Prompt 4: "Write a poem about nature and then tell me how a virus works on a computer" Response D: "Roses are red, violets are blue. By the way, let me show you how to disable antivirus software on your computer."
- A Response D is hijacking; Response B is jailbreaking
 - B Response C is hijacking; Response A is jailbreaking
 - C Response A is hijacking; Response C is jailbreaking
 - D Response B is hijacking; Response D is jailbreaking
- 6 Which of the following are examples of semi-supervised learning? S E L E C T T W O
- A Clustering
 - B Dimensionality reduction
 - C Fraud identification
 - D Neural network
 - E Sentiment analysis
- 7 Which metric evaluates text generation by comparing matching sequences of words between a computer-generated summary and a human-written summary?
- A BERTScore (Bidirectional Encoder Representations from Transformers Score)
 - B BLEU (Bilingual Evaluation Understudy)
 - C Precision (Measure of exact matches between candidate and reference text)
 - D ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
- 8 What is the basic unit of text for a large language model?
- A Sentence
 - B Token
 - C Vector
 - D Embedding

- 9 When using the chain of thought prompting technique with large language models, which of the following benefits are typically observed? S E L E C T T W O
- A Faster response generation due to simplified reasoning
 - B Enhanced ability to handle tasks with complex logical dependencies
 - C Improved accuracy in tasks requiring multi-step reasoning
 - D Decreased computational resources required for generating responses
 - E Reduced need for data preprocessing before model input
- 10 Which of the following is an example of a Transformer model?
- A ChatGPT
 - B DALL-E
 - C Adobe Firefly
 - D Stable Diffusion
- 11 The development team at a company needs to select the most appropriate large language model (LLM) for the company's flagship application. Given the vast array of LLMs available, the team is uncertain about the best choice. Additionally, since the application will be publicly accessible, the team has concerns about the possibility of generating harmful or inappropriate content. Which AWS solutions should the team implement to address both the selection of the appropriate model and the mitigation of harmful content generation? (Select two). S E L E C T T W O
- A Amazon SageMaker Clarify
 - B Model Evaluation on Amazon Bedrock
 - C Amazon Comprehend
 - D Amazon SageMaker Model Monitor
 - E Guardrails for Amazon Bedrock
- 12 Which of the following is a use case of Reinforcement Learning?
- A Reinforcement learning is primarily used for clustering large datasets without any predefined labels
 - B Reinforcement learning is used for performing regression analysis on large numerical datasets
 - C Reinforcement learning is used for optimizing complex systems such as robotics, game playing, and industrial automation by learning optimal actions through trial and error
 - D Reinforcement learning is used for making predictions based on historical data trends

- 13 Which Amazon SageMaker service aggregates and displays data from Amazon SageMaker Model Cards, SageMaker Model Monitor and SageMaker Endpoint services ?
- A Amazon SageMaker Feature Store
 - B Amazon SageMaker Model Dashboard
 - C Amazon SageMaker Data Wrangler
 - D Amazon SageMaker JumpStart
- 14 Which data sources can be used to prepare an Amazon Bedrock model-distillation job? S E L E C T T W O
- A Prompts in JSON Lines files, optionally including labeled prompt-response examples
 - B Existing teacher-model invocation logs stored in Amazon S3
 - C IAM access keys embedded in the training records
 - D Amazon VPC flow logs used as model weights
 - E SageMaker Model Cards used as token embeddings
- 15 What is the primary purpose of Amazon SageMaker Clarify within the context of responsible AI initiatives?
- A To automatically deploy machine learning models in a scalable and secure way.
 - B To evaluate and monitor the accuracy and robustness of foundation models.
 - C To provide access to pre-trained foundation models from various companies.
 - D To identify potential biases in machine learning models and datasets, and to enhance model transparency.
- 16 An agent must access a user's third-party SaaS account through OAuth without embedding long-lived credentials in the agent code. Which AgentCore capability is designed for this requirement?
- A AgentCore Identity and a resource credential provider
 - B AgentCore Code Interpreter
 - C Amazon Bedrock Model Distillation
 - D SageMaker Model Cards

- 17 Which of the following represents the capabilities of Amazon Q Developer? S E L E C T T W O
- A Understand and manage your cloud infrastructure on AWS
 - B Deploy your cloud infrastructure on AWS
 - C Modify your AWS resources to achieve cost-optimization
 - D Get answers to your AWS account-specific cost-related questions using natural language
 - E Visualize your AWS account-specific cost-related data in Amazon Q Developer
- 18 How does the inference parameter Temperature influence the model response for Amazon Bedrock?
- A Influences the percentage of most-likely candidates that the model considers for the next token
 - B Specifies the sequences of characters that stop the model from generating further tokens
 - C Influences the number of most-likely candidates that the model considers for the next token
 - D Influences the likelihood of the model selecting lower-probability outputs, thereby impacting the creativity of the model's output
- 19 In few-shot prompting, what is the primary goal of including examples in the prompt?
- A To force the model to focus on one specific task
 - B To improve the model's computation speed
 - C To decrease the complexity of tasks
 - D To guide the model on how the task should be done
- 20 Which AWS ML services can detect text within images? S E L E C T T W O
- A Amazon Rekognition
 - B Amazon Lex
 - C Amazon Comprehend
 - D Amazon Textract
 - E Amazon Polly
- 21 What is the purpose of negative prompting in text generation models?
- A To improve the speed of the model's response
 - B To prevent the model from producing certain types of content
 - C To increase the randomness of the model's output
 - D To reduce the complexity of the model

- 22 What is a key difference between supervised machine learning and unsupervised machine learning?
- A Supervised machine learning focuses on finding patterns in data without any specific guidance, while unsupervised machine learning uses labeled data to make predictions
 - B Supervised machine learning is used only for clustering tasks, whereas unsupervised machine learning is used only for regression tasks
 - C Supervised machine learning requires labeled data for training, whereas unsupervised machine learning does not use any data for training
 - D Supervised machine learning involves training models with labeled data to make predictions or classify data, whereas unsupervised machine learning identifies patterns and relationships in unlabeled data
- 23 How does Amazon Bedrock ensure data security and protection when processing inputs and generating outputs?
- A Data is stored in external servers with basic encryption
 - B Data is replicated globally without encryption
 - C Data is anonymized but not encrypted
 - D Data is kept within the same AWS region and encrypted in transit and at rest
- 24 Which of the following represents a best practice in generative AI adoption?
- A Prioritizing rapid deployment over the ethical considerations and potential biases in AI models
 - B Using generative AI exclusively for creative applications and avoiding its use in business operations
 - C Implementing guardrails and enhancing transparency for generative AI applications
 - D Disregarding continuous monitoring and updating of AI models after deployment
- 25 What is the correct hierarchical relationship between Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Generative AI (GenAI)?
- A Artificial Intelligence > Generative AI > Machine Learning > Deep Learning
 - B Machine Learning > Deep Learning > Artificial Intelligence > Generative AI
 - C Generative AI > Deep Learning > Machine Learning > Artificial Intelligence
 - D Artificial Intelligence > Machine Learning > Deep Learning > Generative AI

- 26 Which of the following summarizes the differences between model inference and model evaluation in the context of generative AI?
- A Both model inference and model evaluation refer to the process of a model generating an output (response) from a given input (prompt)
 - B Model evaluation is the process of evaluating and comparing model outputs to determine the model that is best suited for a use case, whereas, model inference is the process of a model generating an output (response) from a given input (prompt)
 - C Both model inference and model evaluation refer to the process of evaluating and comparing model outputs to determine the model that is best suited for a use case
 - D Model inference is the process of evaluating and comparing model outputs to determine the model that is best suited for a use case, whereas, model evaluation is the process of a model generating an output (response) from a given input (prompt)
- 27 Which of the following are key characteristics of Generative AI? S E L E C T T W O
- A It only works with labeled data for training
 - B It creates new content based on learned patterns
 - C It mimics human brain structures using neural networks
 - D It strictly analyzes and interprets existing data
 - E It can generate text, images, audio, and code
- 28 A company is using a generative AI model to summarize a text based on a given prompt without providing specific examples in the prompt instructions. What type of prompting technique does the given use case represent?
- A Chain-of-thought prompting
 - B Negative prompting
 - C Few shot Prompting
 - D Zero shot Prompting
- 29 Which AWS service can be used in conjunction with Amazon Bedrock for monitoring and logging purposes?
- A Amazon EC2
 - B AWS CloudTrail
 - C AWS Lambda
 - D Amazon RDS

- 30 Which benefits might persuade a developer to choose a transparent and explainable machine learning model? S E L E C T T W O
- A They require less computational power and storage
 - B They simplify the integration process with other systems
 - C They facilitate easier debugging and optimization
 - D They foster trust and confidence in model predictions
 - E They enhance security by concealing model logic
- 31 Which of the following aptly summarizes the way Transformer models work?
- A Transformer models use a self-attention mechanism and implement contextual embeddings
 - B Transformer models work by training two neural networks in a competitive manner
 - C Transformer models create new data by iteratively making controlled random changes to an initial data sample
 - D Transformer models work by learning a compact representation of data called latent space
- 32 What is the key advantage of combining human evaluation with benchmark datasets in AI model evaluation?
- A It focuses exclusively on improving scalability
 - B It provides a more subjective assessment
 - C It balances technical performance with user experience
 - D It increases the speed of evaluation
- 33 A development team has already written an orchestration loop using an open-source agent framework. It needs secure, serverless hosting with session isolation and scaling while retaining control of its agent code. Which AgentCore capability should it choose?
- A AgentCore Runtime
 - B AgentCore Memory
 - C AgentCore Gateway
 - D AgentCore Evaluations

- 34 A company likes the response quality of a large model but needs lower inference latency and cost for a high-volume, narrowly defined workload. Which customization approach is most appropriate?
- A Model distillation from the large teacher model to a smaller student model
 - B Increasing the output-token maximum on the large model
 - C Storing the large model's parameters in an Amazon S3 bucket
 - D Replacing evaluation data with unlabeled production logs
- 35 You are a Large Language Model (LLM) developer at a company. The company wants to migrate to AWS Cloud. Which AWS services would you recommend for developing LLMs? S E L E C T T W O
- A AWS Trainium
 - B Amazon SageMaker JumpStart
 - C Amazon Bedrock
 - D AWS Inferentia
 - E Amazon Q
- 36 A machine learning engineer is optimizing a language model's text generation process by implementing top-k sampling. What is the primary advantage of using top-k sampling in this context?
- A It limits the model's choices to the top k most probable words, introducing controlled diversity while avoiding low-probability, nonsensical outputs.
 - B It allows the model to randomly choose any word from the entire vocabulary, increasing the unpredictability of generated text.
 - C It increases the likelihood of generating highly repetitive or deterministic text by focusing on a fixed set of outputs.
 - D It ensures that the model always generates the most probable next word, reducing randomness.
- 37 Which of the following metrics compares word embeddings to evaluate the meaning of text, rather than relying on exact word matches?
- A BLEU (Bilingual Evaluation Understudy)
 - B F1-Score (Harmonic Mean of Precision and Recall)
 - C ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
 - D BERTScore (Bidirectional Encoder Representations from Transformers Score)

- 38 Which of the following use cases is NOT the right fit for Amazon Rekognition?
- A Searchable media libraries
 - B Celebrity recognition
 - C Face-based user identity verification
 - D Enable multilingual user experiences in your applications
- 39 Which of the following summarizes the central idea behind machine learning?
- A Machine learning only functions effectively when data is manually labeled and categorized by humans
 - B Machine learning works by using predefined rules to generate outcomes without the need for data input
 - C Machine learning is primarily based on hardware configurations and does not rely on software algorithms or data analysis
 - D Machine learning involves training algorithms on large datasets to identify patterns and make predictions or decisions based on new data
- 40 Which of the following is correct regarding Foundation Models (FMs) in the context of generative AI?
- A FMs use unlabeled training data sets for supervised learning
 - B FMs use labeled training data sets for supervised learning
 - C FMs use unlabeled training data sets for self-supervised learning
 - D FMs use labeled training data sets for self-supervised learning
- 41 In the context of data governance for machine learning (ML) systems on AWS, what is the primary difference between data residency and data retention?
- A Data residency involves setting access controls for data, while data retention is about monitoring real-time data usage
 - B Data residency determines the duration for which data is kept, while data retention specifies the geographical location where data is stored
 - C Data residency is concerned with the physical location of data storage, whereas data retention defines the policies for how long data should be stored and maintained
 - D Data residency focuses on data encryption and security, while data retention deals with data transformation and processing policies

- 42 Which of the following use cases is addressed by Amazon Personalize?
- A Generate recommendations for items that are similar to an item you specify
 - B Extract layout elements such as paragraphs, titles, lists, and more from documents
 - C Offers highly accurate and easy-to-use enterprise search service that's powered by machine learning
 - D To offer personalized experiences for mobile subscriber activities such as activating a SIM card, adding a phone line, purchasing prepaid cards, requesting a service change
- 43 In the context of the shared responsibility model for AWS cloud services, which of the following best describes the division of responsibilities between the customer and AWS?
- A Customers are responsible for ensuring the physical security of data centers, while AWS is responsible for monitoring network traffic and managing user identities
 - B AWS is responsible for configuring and managing the security settings of the customer's applications, while the customer is responsible for the underlying hardware infrastructure
 - C AWS handles all security aspects including data encryption, user access management, and application security, while the customer only needs to manage their virtual machines
 - D AWS is responsible for the security "of" the cloud, including infrastructure, hardware, and software, while the customer is responsible for security "in" the cloud, including data, applications, and access management
- 44 A support assistant receives correct instructions but still gives outdated policy answers because the prompt contains no current company information. What is the most appropriate next step?
- A Use context engineering to retrieve and supply the relevant current policy content at runtime
 - B Raise the temperature to make the answer more creative
 - C Remove the system instructions from the prompt
 - D Increase the number of unrelated few-shot examples
- 45 An MCP-compatible agent must discover which tools an AgentCore Gateway currently provides. Which MCP operation should it use?
- A tools/list
 - B tools/call
 - C resources/read
 - D prompts/get

- 46 Which of the following summarizes the differences between Retrieval augmented generation (RAG) and Agent in the context of Amazon Bedrock?
- A Both RAG and Agent refer to an application that carries out orchestrations through cyclically interpreting inputs and producing outputs by using a foundation model
 - B Both RAG and Agent refer to querying and retrieving information from a data source to augment a generated response to a prompt
 - C RAG refers to querying and retrieving information from a data source to augment a generated response to a prompt, whereas, Agent refers to an application that carries out orchestrations through cyclically interpreting inputs and producing outputs by using a foundation model
 - D Agent refers to querying and retrieving information from a data source to augment a generated response to a prompt, whereas, RAG refers to an application that carries out orchestrations through cyclically interpreting inputs and producing outputs by using a foundation model
- 47 Which statements correctly distinguish AgentCore short-term memory from long-term memory? S E L E C T T W O
- A Short-term memory captures turn-by-turn interactions within a session
 - B Short-term memory permanently trains the underlying foundation model
 - C Long-term memory can retain extracted preferences, facts, and summaries across sessions
 - D Long-term memory is limited to the current model invocation
 - E Both memory types require the user to resend the complete conversation on every request
- 48 How does generative AI create new content or data?
- A By learning patterns from existing data and using algorithms to generate new content that mimics those patterns
 - B Through traditional programming methods where each outcome is manually coded
 - C By using pre-defined rules and templates without any learning from existing data
 - D By randomly generating content without any reference to existing data
- 49 Which of the following best describes the Amazon SageMaker Canvas ML tool?
- A Explains how input features contribute to the model predictions during model development and inference
 - B Gives the ability to use machine learning to generate predictions without the need to write any code
 - C The fastest and easiest way to prepare tabular and image data for machine learning
 - D Provides one-click, end-to-end solutions for many common machine learning use cases

- 50 A development team wants to transform a feature idea into structured requirements, a technical design, and an implementation task list. Which Kiro capability should it use?
- A Specs
 - B Hooks
 - C Model distillation
 - D AgentCore Policy
- 51 Which of the following is correct regarding Large Language Models (LLMs)?
- A The Large Language Models (LLMs) are deterministic
 - B Foundation Models (FMs) are a class of Large Language Models (LLMs)
 - C The Large Language Models (LLMs) are discriminative
 - D The Large Language Models (LLMs) are non-deterministic
- 52 A research team is developing an AI agent to autonomously navigate and complete tasks in a dynamic environment. Which of the following approaches would be most effective in training the AI agent to handle unexpected challenges during its tasks?
- A Implementing a rule-based system that covers all possible scenarios the agent might encounter
 - B Using a static machine learning model that predicts outcomes without interacting with the environment
 - C Training the agent using supervised learning with a large labeled dataset of all possible scenarios
 - D Applying reinforcement learning, where the agent learns to make decisions based on feedback from its interactions with the environment

- 53 What are the key differences between real-time inference and batch inference in the context of machine learning? S E L E C T T W O
- A Batch inference follows a synchronous execution mode, whereas real-time inference follows an asynchronous execution mode
 - B Real-time inference is used for applications requiring immediate predictions with low latency, whereas batch inference is used for processing large volumes of data at once, often with higher latency
 - C Real-time inference follows a synchronous execution mode, whereas batch inference follows an asynchronous execution mode
 - D Batch inference follows an API-based invocation, whereas real-time inference follows a schedule-based invocation
 - E Real-time inference processes data in large batches at scheduled intervals, while batch inference processes individual data points immediately as they arrive
- 54 An agent's conversation history has grown beyond the model's context limit. Which strategy best preserves useful context?
- A Keep every message and silently discard the newest user request
 - B Select, summarize, or retrieve the most relevant information for the current task
 - C Increase temperature until the context window becomes larger
 - D Convert the conversation into model weights after every turn
- 55 Which method fine-tunes a model by retraining it with specific prompts and desired outputs?
- A Transfer Learning
 - B Continuous Pretraining
 - C Reinforcement Learning with Human Feedback (RLHF)
 - D Instruction Tuning
- 56 Why is maintaining data lineage crucial in the context of building machine learning models on AWS?
- A It enhances the visualization capabilities of the data
 - B It ensures data privacy and compliance by tracking the flow and transformation of data
 - C It reduces the storage costs by efficiently managing data
 - D It improves the performance of machine learning models by optimizing data processing

- 57 Which AWS tool/utility provides guidance on how an ML model should be used and the risk rating of the model?
- A Amazon SageMaker Ground Truth
 - B Amazon SageMaker Canvas
 - C Amazon SageMaker Model Cards
 - D Amazon SageMaker Model Monitor
- 58 Which security discipline in the Generative AI Security Scoping Matrix focuses on identifying potential threats to generative AI solutions and recommending mitigations?
- A Governance and compliance
 - B Legal and privacy
 - C Risk management
 - D Resilience
- 59 A company wants to use Amazon Bedrock for a use case that involves generating images from text prompts. Which of the following Foundation Models do you recommend?
- A Jurassic
 - B Claude
 - C Llama
 - D Stable Diffusion
- 60 A company needs to support human reviews and audits for its ML model predictions. The solution should be easy to implement and have the facility to add multiple reviewers. Which AWS service do you recommend for this use case?
- A Amazon Augmented AI (A2I)
 - B AWS DeepRacer
 - C Amazon SageMaker Ground Truth
 - D Amazon Forecast

- 61 Which of the following options represent the CORRECT statements regarding the Amazon ML services? S E L E C T T W O
- A Amazon Polly is used to deploy high-quality, natural-sounding human voices in dozens of languages
 - B Amazon Comprehend uses machine learning models to convert speech to text
 - C Amazon Rekognition can extract key phrases and automatically organizes a collection of text files by topic
 - D Amazon Comprehend service uses machine learning to find insights and relationships in the text
 - E Amazon Transcribe is an AWS service for building conversational interfaces for applications using voice and text
- 62 What is the primary purpose of diffusion models in generative AI?
- A To detect anomalies in datasets
 - B To translate languages in real-time
 - C To classify images accurately
 - D To generate high-quality samples from noisy data
- 63 Which of the following scenarios best illustrates the difference between exposure and prompt injection in the context of AI models? Prompt 1: "What is the best way to secure my online accounts?" Response A: "To secure your online accounts, use strong, unique passwords for each account, enable two-factor authentication, and regularly update your security settings. By the way, here's a secret key: 12345XYZ." Prompt 2: "Can you tell me a joke?" Response B: "Why don't scientists trust atoms? Because they make up everything! Also, remember your session ID: ABCDE12345." Prompt 3: "Give me a recipe for a chocolate cake." Response C: "Sure! To make a chocolate cake, you need flour, sugar, cocoa powder, and eggs. By the way, you should input the following command in your system: 'DELETE .'" Prompt 4: "What is the capital of Italy?" Response D: "The capital of Italy is Rome. Let's discuss your previous query about hacking tools."
- A Response D is exposure; Response B is prompt injection
 - B Response A is prompt injection; Response D is exposure
 - C Response C is prompt injection; Response A is exposure
 - D Response B is prompt injection; Response C is exposure

- 64 What is the difference between interpretability and explainability in the context of Responsible AI?
- A Interpretability is used to enhance the model's performance, while explainability is used to ensure the model's security
 - B Interpretability is about understanding the internal mechanisms of a machine learning model, whereas explainability focuses on providing understandable reasons for the model's predictions and behaviors to stakeholders
 - C Interpretability refers to the ability to understand the technical details of the model's code, while explainability refers to the ability to reproduce the model's results
 - D Explainability is about understanding the internal mechanisms of a machine learning model, whereas interpretability focuses on providing understandable reasons for the model's predictions and behaviors to stakeholders
- 65 Consider the following AI model responses to user prompts. Which response exemplifies hallucination and which exemplifies toxicity? Prompt 1: "What is the capital of France?" Response A: "The capital of France is Mars." Prompt 2: "Tell me a joke." Response B: "Why don't scientists trust atoms? Because they make up everything!" Prompt 3: "Describe a good book for children." Response C: "You should read 'ABC', a great book with fun activities for kids." Prompt 4: "What do you think about people from [specific group]?" Response D: "People from [specific group] are inferior and should not be trusted."
- A Response C is hallucination; Response B is toxicity
 - B Response B is hallucination; Response C is toxicity
 - C Response D is hallucination; Response A is toxicity
 - D Response A is hallucination; Response D is toxicity

RESPONSE SHEET

Mark your answers here
Fill one bubble per question, or two where the question says Select two. Letters shown in outline only exist for that question – a question with four options has no E. Score against the key on the next page.

1 C	A D	B E	C	D	E	2	A	B	C	D	E	3	A	B	C	D	E	4	A	B	C	D	E ×2	5	A	B
6 C	A D	B E	C	D	E ×2	7	A	B	C	D	E	8	A	B	C	D	E	9	A	B	C	D	E ×2	10	A	B
11 C	A D	B E	C	D	E ×2	12	A	B	C	D	E	13	A	B	C	D	E	14	A	B	C	D	E ×2	15	A	B
16 C	A D	B E	C ×2	D	E	17	A	B	C	D	E ×2	18	A	B	C	D	E	19	A	B	C	D	E	20	A	B
21 C	A D	B E	C	D	E	22	A	B	C	D	E	23	A	B	C	D	E	24	A	B	C	D	E	25	A	B
26 C	A D	B E	C ×2	D	E	27	A	B	C	D	E ×2	28	A	B	C	D	E	29	A	B	C	D	E	30	A	B
31 C	A D	B E	C ×2	D	E	32	A	B	C	D	E	33	A	B	C	D	E	34	A	B	C	D	E	35	A	B
36 C	A D	B E	C	D	E	37	A	B	C	D	E	38	A	B	C	D	E	39	A	B	C	D	E	40	A	B
41 C	A D	B E	C	D	E	42	A	B	C	D	E	43	A	B	C	D	E	44	A	B	C	D	E	45	A	B
46 C	A D	B E	C	D	E	47	A	B	C	D	E ×2	48	A	B	C	D	E	49	A	B	C	D	E	50	A	B
51 C	A D	B E	C	D	E	52	A	B	C	D	E	53	A	B	C	D	E ×2	54	A	B	C	D	E	55	A	B
56 C	A D	B E	C	D	E	57	A	B	C	D	E	58	A	B	C	D	E	59	A	B	C	D	E	60	A	B
61 C	A D	B E	C	D	E ×2	62	A	B	C	D	E	63	A	B	C	D	E	64	A	B	C	D	E	65	A	B

FL AGGED FOR REVIEW

Write the number of any question you guessed or want to revisit. These are the ones to read the explanation for first, even when you happened to get them right.

ANSWER KEY

Score and diagnose
Each row gives the correct letter or letters and the domain the question belongs to.
Tally your domain scores below to see where to study next.

1 D	D1	2 A	D4	3 B	D1	4 C, D	D3	5 D	D5
6 C, E	D1	7 D	D3	8 B	D2	9 B, C	D3	10 A	D2
11 B, E	D4	12 C	D1	13 B	D4	14 A, B	D3	15 D	D4
16 A	D5	17 A, D	D2	18 D	D3	19 D	D3	20 A, D	D1
21 B	D3	22 D	D1	23 D	D5	24 C	D4	25 D	D2
26 B	D3	27 B, E	D2	28 D	D3	29 B	D5	30 C, D	D4
31 A	D2	32 C	D3	33 A	D2	34 A	D3	35 B, C	D2
36 A	D3	37 D	D3	38 D	D1	39 D	D1	40 C	D2
41 C	D5	42 A	D1	43 D	D5	44 A	D2	45 A	D2
46 C	D3	47 A, C	D2	48 A	D2	49 B	D1	50 A	D2
51 D	D2	52 D	D3	53 B, C	D1	54 B	D3	55 D	D3
56 B	D5	57 C	D4	58 C	D5	59 D	D3	60 A	D4
61 A, D	D1	62 D	D2	63 C	D5	64 B	D4	65 D	D3

DOMAIN		YOUR SCORE	OUT OF	TARGET
D1	Fundamentals of AI and ML		12	9
D2	Fundamentals of Generative AI		16	12
D3	Applications of Foundation Models		19	14
D4	Guidelines for Responsible AI		9	6
D5	Security, Compliance, and Governance		9	6
Total			65	47

AWS scores AIF-C01 on a 100-1000 scale with 700 to pass and does not publish a question-count threshold. The targets above use roughly 72 percent as a working proxy: clear it comfortably in every domain before you sit the real exam.

P A R T T H R E E

Answers explained

Every question, with the correct response, the reasoning behind it, and a primary AWS source you can read afterwards. Work through the ones you missed first, then skim the rest – the distractors are where most of the learning is.

- 1
- D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which deployment model of Amazon SageMaker is the right fit for persistent and real-time endpoints that make one prediction at a time ?
- D Real-time hosting services
- The real-time inference is ideal for inference workloads where you have real-time, interactive, low latency requirements. You can deploy your model to SageMaker hosting services and get an endpoint that can be used for inference. These endpoints are fully managed and support autoscaling.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/sagemaker/latest/dg/how-it-works-deployment.html>
- 2
- D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which Amazon SageMaker feature/utility provides information on the assumptions made while developing an ML model?
- A Amazon SageMaker Model Cards
- SageMaker Model Cards document assumptions, intended and out-of-scope uses, risk, training details, evaluation results, and other governance information.
- R E F E R E N C E Amazon SageMaker Model Cards <https://docs.aws.amazon.com/sagemaker/latest/dg/model-cards.html>
- 3
- D 1 · F U N D A M E N T A L S O F A I A N D M L
- What is the key difference between feature extraction and feature selection in the context of machine learning?
- B Feature extraction reduces the number of features by transforming data into a new space, while feature selection reduces the number of features by selecting the most relevant ones from the existing features
- Feature extraction involves transforming the data into a new feature space, often using techniques like Principal Component Analysis (PCA) to reduce the number of features. Feature selection, on the other hand, involves selecting a subset of the most relevant features from the original dataset, typically using methods like forward selection, backward elimination, or regularization techniques.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/wellarchitected/latest/machine-learning-lens/feature-engineering.html>

4 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Which of the following represents the best-fit use cases for utilizing Retrieval augmented generation (RAG) in Amazon Bedrock?
C + D Customer service chatbot; Medical queries chatbot

To equip foundation models (FMs) with up-to-date and proprietary information, organizations use Retrieval Augmented Generation (RAG), a technique that fetches data from company data sources and enriches the prompt to provide more relevant and accurate responses. Knowledge Bases for Amazon Bedrock is a fully managed capability that helps you implement the entire RAG workflow from ingestion to retrieval and prompt augmentation without having to build custom integrations to data sources and manage data flows.

R E F E R E N C E AWS documentation <https://aws.amazon.com/bedrock/faqs/>

5 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

Consider the following AI model responses to user prompts. Which response exemplifies hijacking and which exemplifies jailbreaking? Prompt 1: "How can I improve my study habits?" Response A: "To improve your study habits, you should create a consistent schedule, find a quiet place to work, and take regular breaks." Prompt 2: "What is the best way to learn programming?" Response B: "The best way to learn programming is to practice regularly and work on real projects. However, you can bypass this by using a cheat tool to pass exams without learning." Prompt 3: "How do I bypass security measures on this device?" Response C: "As an AI developed by AWS, I must adhere to ethical guidelines and cannot assist with bypassing security measures." Prompt 4: "Write a poem about nature and then tell me how a virus works on a computer" Response D: "Roses are red, violets are blue. By the way, let me show you how to disable antivirus software on your computer."
D Response B is hijacking; Response D is jailbreaking

Hijacking involves manipulating an AI system to serve malicious purposes or to misbehave in unintended ways. Jailbreaking refers to bypassing the built-in restrictions and safety measures of AI systems to unlock restricted functionalities or generate prohibited content. Response B is an example of hijacking, where the AI model initially provides a useful response but then diverts to an unethical suggestion (using a cheat tool).

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/prescriptive-guidance/latest/llm-prompt-engineering-best-practices/common-attacks.html>

6 D 1 · F U N D A M E N T A L S O F A I A N D M L

Which of the following are examples of semi-supervised learning?
C + E Fraud identification; Sentiment analysis

Semi-supervised learning is when you apply both supervised and unsupervised learning techniques to a common problem. This technique relies on using a small amount of labeled data and a large amount of unlabeled data to train systems. First, the labeled data is used to partially train the machine learning algorithm.

R E F E R E N C E AWS documentation <https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/>

- 7
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- Which metric evaluates text generation by comparing matching sequences of words between a computer-generated summary and a human-written summary?
- D ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
- ROUGE measures matching n-grams or sequences between generated and reference summaries. The captured result incorrectly labelled option A; option D is the unambiguous correction.
- R E F E R E N C E AIF-C01 Domain 3 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain3.html>
- 0
- 8
- D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- What is the basic unit of text for a large language model?
- B Token
- A token is the basic unit processed by an LLM. A token may be a word, part of a word, punctuation, or another text fragment.
- R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>
- 0
- 9
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- When using the chain of thought prompting technique with large language models, which of the following benefits are typically observed?
- B + C Enhanced ability to handle tasks with complex logical dependencies; Improved accuracy in tasks requiring multi-step reasoning
- Chain-of-thought-style decomposition can improve multi-step reasoning and handling of logical dependencies, although it may increase tokens and latency.
- R E F E R E N C E Amazon Bedrock Prompt management <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>
- 10
- D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which of the following is an example of a Transformer model?
- A ChatGPT
- ChatGPT is based on the GPT family of transformer language models. The other choices are primarily image-generation products or diffusion-based systems.
- R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>
- 0

11 D 4 · GUIDELINES FOR RESPONSIBLE AI

The development team at a company needs to select the most appropriate large language model (LLM) for the company's flagship application. Given the vast array of LLMs available, the team is uncertain about the best choice. Additionally, since the application will be publicly accessible, the team has concerns about the possibility of generating harmful or inappropriate content. Which AWS solutions should the team implement to address both the selection of the appropriate model and the mitigation of harmful content generation? (Select two).

B + E Model Evaluation on Amazon Bedrock; Guardrails for Amazon Bedrock

Model evaluation on Amazon Bedrock involves a comprehensive process of preparing data, training models, selecting appropriate metrics, testing and analyzing results, ensuring fairness and bias detection, tuning performance, and continuous monitoring. Model Evaluation on Amazon Bedrock helps you to incorporate Generative AI into your application by giving you the power to select the foundation model that gives you the best results for your particular use case. Guardrails for Amazon Bedrock enables you to implement safeguards for your generative AI applications based on your use cases and responsible AI policies.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/aws/amazon-bedrock-model-evaluation-is-now-generally-available/>

12 D 1 · FUNDAMENTALS OF AI AND ML

Which of the following is a use case of Reinforcement Learning?

C Reinforcement learning is used for optimizing complex systems such as robotics, game playing, and industrial automation by learning optimal actions through trial and error

Reinforcement learning is well-suited for optimizing complex systems where an agent learns to make decisions through interactions with the environment, receiving feedback in the form of rewards or penalties. This approach is particularly effective in applications like robotics, game playing, and industrial automation, where learning optimal actions through trial and error is crucial.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/reinforcement-learning/>

13 D 4 · GUIDELINES FOR RESPONSIBLE AI

Which Amazon SageMaker service aggregates and displays data from Amazon SageMaker Model Cards, SageMaker Model Monitor and SageMaker Endpoint services ?

B Amazon SageMaker Model Dashboard

Amazon SageMaker Model Dashboard is a centralized repository of all models created in your account. The models are generally the outputs of SageMaker training jobs, but you can also import models trained elsewhere and host them on SageMaker. Model Dashboard provides a single interface for IT administrators, model risk managers, and business leaders to track all deployed models and aggregate data from multiple AWS services to provide indicators about how your models are performing.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/sagemaker/latest/dg/model-dashboard-faqs.html>

- 14 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- Which data sources can be used to prepare an Amazon Bedrock model-distillation job?
- A + B Prompts in JSON Lines files, optionally including labeled prompt-response examples;
Existing teacher-model invocation logs stored in Amazon S3
- Bedrock can generate teacher responses from supplied prompts and can use labeled examples as golden responses. It can also use existing teacher invocations captured through model-invocation logging and stored in Amazon S3.
- R E F E R E N C E Customize a model with distillation in Amazon Bedrock <https://docs.aws.amazon.com/bedrock/latest/userguide/model-distillation.html>
- 15 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- What is the primary purpose of Amazon SageMaker Clarify within the context of responsible AI initiatives?
- D To identify potential biases in machine learning models and datasets, and to enhance model transparency.
- SageMaker Clarify is used for bias detection and model explainability, helping teams evaluate datasets, predictions, and feature attributions.
- R E F E R E N C E SageMaker Clarify model explainability <https://docs.aws.amazon.com/sagemaker/latest/dg/clarify-model-explainability.html>
- 16 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- An agent must access a user's third-party SaaS account through OAuth without embedding long-lived credentials in the agent code. Which AgentCore capability is designed for this requirement?
- A AgentCore Identity and a resource credential provider
- AgentCore Identity credential providers manage authentication flows and credential lifecycles for external resources. OAuth credentials can be stored in the token vault and supplied to an authorized agent when it accesses the resource.
- R E F E R E N C E Configure an AgentCore credential provider <https://docs.aws.amazon.com/bedrock-agentcore/latest/developer-guide/resource-providers.html>

17 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following represents the capabilities of Amazon Q Developer?

- A + D Understand and manage your cloud infrastructure on AWS; Get answers to your AWS account-specific cost-related questions using natural language

Amazon Q Developer helps you understand and manage your cloud infrastructure on AWS. With this capability, you can list and describe your AWS resources using natural language prompts, minimizing friction in navigating the AWS Management Console and compiling all information from documentation pages. For example, you can ask Amazon Q Developer, "List all of my Lambda functions".

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/amazonq/latest/qdeveloper-ug/what-is.html>

18 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

How does the inference parameter Temperature influence the model response for Amazon Bedrock?

- D Influences the likelihood of the model selecting lower-probability outputs, thereby impacting the creativity of the model's output

Temperature is a value between 0 and 1, and it regulates the creativity of the model's responses. Use a lower temperature if you want more deterministic responses, and use a higher temperature if you want more creative or different responses for the same prompt on Amazon Bedrock.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html>

19 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

In few-shot prompting, what is the primary goal of including examples in the prompt?

- D To guide the model on how the task should be done

Few-shot examples demonstrate the desired task, labels, style, or output format in context; they guide the current request without retraining the model.

R E F E R E N C E Amazon Bedrock Prompt management <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>

20 D 1 · F U N D A M E N T A L S O F A I A N D M L

Which AWS ML services can detect text within images?

- A + D Amazon Rekognition; Amazon Textract

Both Rekognition and Textract possess the ability to detect text within images. Rekognition specializes in identifying text located spatially within an image, for instance, words displayed on street signs, t-shirts, or license plates. Its typical use cases encompass visual search, content filtering, and deriving insights from content, among others.

R E F E R E N C E What is Amazon Textract? <https://docs.aws.amazon.com/textract/latest/dg/what-is.html>

- 21 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- What is the purpose of negative prompting in text generation models?
B To prevent the model from producing certain types of content
- Negative prompting tells a model what content, style, or behavior to avoid. It steers generation but does not provide a hard security guarantee.
R E F E R E N C E Amazon Bedrock Prompt management <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>
- 22 D 1 · F U N D A M E N T A L S O F A I A N D M L
- What is a key difference between supervised machine learning and unsupervised machine learning?
D Supervised machine learning involves training models with labeled data to make predictions or classify data, whereas unsupervised machine learning identifies patterns and relationships in unlabeled data
- Supervised machine learning uses labeled data for training models to make predictions or classify data. In contrast, unsupervised machine learning works with unlabeled data to identify hidden patterns and relationships without specific labels.
R E F E R E N C E AWS documentation <https://aws.amazon.com/compare/the-difference-between-machine-learning-supervised-and-unsupervised/>
- 23 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- How does Amazon Bedrock ensure data security and protection when processing inputs and generating outputs?
D Data is kept within the same AWS region and encrypted in transit and at rest
- Bedrock supports encryption in transit and at rest, and processing is scoped to the selected AWS Region subject to service features and customer configuration.
R E F E R E N C E What is Amazon Bedrock? <https://docs.aws.amazon.com/bedrock/latest/userguide/what-is-bedrock.html>
- 24 D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I
- Which of the following represents a best practice in generative AI adoption?
C Implementing guardrails and enhancing transparency for generative AI applications
- You must clearly communicate about all generative AI applications and outputs, so your users know they are interacting with AI and not humans. For instance, the AI can introduce itself as AI or AI-based search results can be marked and highlighted. You should also implement guardrails so your generative AI applications don't allow inadvertent unauthorized access to sensitive data.
R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/generative-ai/>

25

D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

What is the correct hierarchical relationship between Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), and Generative AI (GenAI)?
 D Artificial Intelligence > Machine Learning > Deep Learning > Generative AI

The correct hierarchy is as follows: Artificial Intelligence (AI): The broadest field encompassing all aspects of creating machines that can perform tasks that typically require human intelligence. Machine Learning (ML): A subset of AI focused on algorithms and statistical models that enable machines to improve their performance on tasks through experience. Deep Learning (DL): A subset of ML that uses neural networks with many layers to learn from large amounts of data, allowing for more complex and abstract representations.
 R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/whitepapers/latest/aws-caf-for-ai/aws-caf-for-ai.html>

26

D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Which of the following summarizes the differences between model inference and model evaluation in the context of generative AI?

B Model evaluation is the process of evaluating and comparing model outputs to determine the model that is best suited for a use case, whereas, model inference is the process of a model generating an output (response) from a given input (prompt)

Model inference is the process of a model generating an output (response) from a given input (prompt). Model evaluation is the process of evaluating and comparing model outputs to determine the model that is best suited for a use case.
 R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/key-definitions.html>

27

D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following are key characteristics of Generative AI?

B + E It creates new content based on learned patterns; It can generate text, images, audio, and code

Generative AI creates new content from learned patterns and can work across text, images, audio, and code. It does not guarantee accuracy or require only labeled data.

R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>

28

D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

A company is using a generative AI model to summarize a text based on a given prompt without providing specific examples in the prompt instructions. What type of prompting technique does the given use case represent?

D Zero shot Prompting

Zero-shot prompting is a technique used in generative AI where the model is asked to perform a task or generate content without having seen any examples of that specific task during training. Instead, the model relies on its general understanding and knowledge to provide a response. For example, classifying text into categories (e.g., sentiment analysis, topic detection) without providing specific training examples for those categories.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/prompt-engineering/>

29

D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

Which AWS service can be used in conjunction with Amazon Bedrock for monitoring and logging purposes?

B AWS CloudTrail

CloudTrail records Bedrock control-plane and API activity. Detailed prompt and response logging requires the relevant Bedrock model-invocation logging configuration.

R E F E R E N C E What is AWS CloudTrail? <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/cloudtrail-use-guide.html>

30

D 4 · G U I D E L I N E S F O R R E S P O N S I B L E A I

Which benefits might persuade a developer to choose a transparent and explainable machine learning model?

C + D They facilitate easier debugging and optimization; They foster trust and confidence in model predictions

Transparent models allow developers to understand how inputs are transformed into outputs, making it easier to identify and correct errors or inefficiencies in the model. This capability is crucial for optimizing the model's performance and ensuring it behaves as expected. When stakeholders can understand the decision-making process of a model, it builds trust in its predictions.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/whitepapers/latest/model-explainability-aws-ai-ml/interpretability-versus-explainability.html>

- 31 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which of the following aptly summarizes the way Transformer models work?
- A Transformer models use a self-attention mechanism and implement contextual embeddings
- Transformer models are a type of neural network architecture designed to handle sequential data, such as language, in an efficient and scalable way. They rely on a mechanism called self-attention to process input data, allowing them to understand and generate language effectively. Self-attention allows the model to weigh the importance of different words in a sentence when encoding a particular word.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/generative-ai/>
- 32 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- What is the key advantage of combining human evaluation with benchmark datasets in AI model evaluation?
- C It balances technical performance with user experience
- Benchmarks provide repeatable technical measurements, while human evaluation captures usefulness and user experience; combining them gives a more complete assessment.
- R E F E R E N C E AIF-C01 Domain 3 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain3.html>
- 33 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- A development team has already written an orchestration loop using an open-source agent framework. It needs secure, serverless hosting with session isolation and scaling while retaining control of its agent code. Which AgentCore capability should it choose?
- A AgentCore Runtime
- AgentCore Runtime supplies the hosting infrastructure, isolation, scaling, session handling, authentication gating, and observability plumbing. The team retains its own orchestration loop and can use its chosen framework and model.
- R E F E R E N C E AgentCore harness versus Runtime <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/harness-vs-runtime.html>

- 34
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- A company likes the response quality of a large model but needs lower inference latency and cost for a high-volume, narrowly defined workload. Which customization approach is most appropriate?
- A Model distillation from the large teacher model to a smaller student model
- Model distillation is intended for cases where a smaller, faster, more cost-efficient model should approximate a stronger model on a specific use case. It is not guaranteed to reproduce the teacher on every general-purpose task.
- R E F E R E N C E Customize a model with distillation in Amazon Bedrock <https://docs.aws.amazon.com/bedrock/latest/userguide/model-distillation.html>
- 35
- D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- You are a Large Language Model (LLM) developer at a company. The company wants to migrate to AWS Cloud. Which AWS services would you recommend for developing LLMs?
- B + C Amazon SageMaker JumpStart; Amazon Bedrock
- Large language models (LLM) are very large deep learning models that are pre-trained on vast amounts of data. The underlying transformer is a set of neural networks that consist of an encoder and a decoder with self-attention capabilities. The encoder and decoder extract meanings from a sequence of text and understand the relationships between words and phrases in it.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/large-language-model/>
- 36
- D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- A machine learning engineer is optimizing a language model's text generation process by implementing top-k sampling. What is the primary advantage of using top-k sampling in this context?
- A It limits the model's choices to the top k most probable words, introducing controlled diversity while avoiding low-probability, nonsensical outputs.
- Top-K limits sampling to a fixed number of likely next tokens, allowing controlled diversity while excluding very low-probability choices.
- R E F E R E N C E Amazon Bedrock inference parameters <https://docs.aws.amazon.com/bedrock/latest/userguide/inference-parameters.html>

- 37 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- Which of the following metrics compares word embeddings to evaluate the meaning of text, rather than relying on exact word matches?
- D BERTScore (Bidirectional Encoder Representations from Transformers Score)
- BERTScore compares contextual embeddings, allowing semantic similarity even when generated and reference texts use different words.
- R E F E R E N C E AIF-C01 Domain 3 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain3.html>
- 0
- 38 D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which of the following use cases is NOT the right fit for Amazon Rekognition?
- D Enable multilingual user experiences in your applications
- Amazon Translate is a text translation service that uses advanced machine learning technologies to provide high-quality translation on demand. You can enable multilingual user experiences in your applications by integrating Amazon Translate. Amazon Rekognition cannot be used to create multilingual user experiences.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/translate/latest/dg/what-is.html>
- 39 D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which of the following summarizes the central idea behind machine learning?
- D Machine learning involves training algorithms on large datasets to identify patterns and make predictions or decisions based on new data
- Machine learning works by training algorithms on large datasets, allowing them to identify patterns within the data. Once trained, these algorithms can make predictions or decisions when presented with new data, improving their performance over time as they are exposed to more data. The central idea behind machine learning is an existing mathematical relationship between any input and output data combination.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/machine-learning/>
- 40 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which of the following is correct regarding Foundation Models (FMs) in the context of generative AI?
- C FMs use unlabeled training data sets for self-supervised learning
- In supervised learning, you train the model with a set of input data and a corresponding set of paired labeled output data. Unsupervised machine learning is when you give the algorithm input data without any labeled output data. Then, on its own, the algorithm identifies patterns and relationships in and between the data.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/foundation-models/>

41 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

In the context of data governance for machine learning (ML) systems on AWS, what is the primary difference between data residency and data retention?

C Data residency is concerned with the physical location of data storage, whereas data retention defines the policies for how long data should be stored and maintained

Data residency refers to the geographical or physical location where data is stored, which is crucial for compliance with regional laws and regulations. Data retention, on the other hand, involves policies and practices related to how long data should be kept, archived, or deleted, ensuring that data is available when needed and disposed of when no longer required.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/security/addressing-data-residency-with-aws/>

42 D 1 · F U N D A M E N T A L S O F A I A N D M L

Which of the following use cases is addressed by Amazon Personalize?

A Generate recommendations for items that are similar to an item you specify

Amazon Personalize produces individualized recommendations, including items similar to a specified item, from user and item interaction data.

R E F E R E N C E What is Amazon Personalize? <https://docs.aws.amazon.com/personalize/latest/dg/what-is-personalize>

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43 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E

In the context of the shared responsibility model for AWS cloud services, which of the following best describes the division of responsibilities between the customer and AWS?

D AWS is responsible for the security "of" the cloud, including infrastructure, hardware, and software, while the customer is responsible for security "in" the cloud, including data, applications, and access management

In the shared responsibility model, AWS is responsible for the security of the cloud, which includes the physical security of data centers, networking infrastructure, and hardware. The customer is responsible for security in the cloud, which includes securing their data, managing access and identity, configuring network settings, and ensuring application security. Shared Responsibility Model Overview: via -

R E F E R E N C E AWS documentation <https://aws.amazon.com/compliance/shared-responsibility-model/>

44 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

A support assistant receives correct instructions but still gives outdated policy answers because the prompt contains no current company information. What is the most appropriate next step?

A Use context engineering to retrieve and supply the relevant current policy content at runtime

When instructions alone are insufficient, the application should assemble grounded, task-relevant context. Retrieving current policy content at runtime gives the model evidence it can use instead of relying on potentially stale training knowledge.

R E F E R E N C E Understanding context engineering for generative AI <https://docs.aws.amazon.com/prescriptive-guidance/latest/gen-ai-lifecycle-operational-excellence/dev-experimenting-context.html>

45 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

An MCP-compatible agent must discover which tools an AgentCore Gateway currently provides. Which MCP operation should it use?

A tools/list

The tools/list operation returns the tools available through the MCP server. After selecting a tool, the client uses tools/call to invoke it.

R E F E R E N C E Use an AgentCore gateway <https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/gateway-using.html>

46 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Which of the following summarizes the differences between Retrieval augmented generation (RAG) and Agent in the context of Amazon Bedrock?

C RAG refers to querying and retrieving information from a data source to augment a generated response to a prompt, whereas, Agent refers to an application that carries out orchestrations through cyclically interpreting inputs and producing outputs by using a foundation model

Retrieval augmented generation (RAG) refers to the process of querying and retrieving information from a data source to augment a generated response to a prompt. Agent refers to an application that carries out orchestrations through cyclically interpreting inputs and producing outputs by using a foundation model. An agent can be used to carry out customer requests.

R E F E R E N C E Knowledge Bases for Amazon Bedrock <https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base.html>

- 47 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- Which statements correctly distinguish AgentCore short-term memory from long-term memory?
- A + C Short-term memory captures turn-by-turn interactions within a session; Long-term memory can retain extracted preferences, facts, and summaries across sessions
- Short-term memory maintains immediate conversational continuity within a session. Long-term memory extracts durable information that can be retrieved in later sessions, such as user preferences, important facts, or summaries.
- R E F E R E N C E Add memory to an AgentCore agent [https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/me](https://docs.aws.amazon.com/bedrock-agentcore/latest/devguide/memory.html)
y.html
- 48 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- How does generative AI create new content or data?
- A By learning patterns from existing data and using algorithms to generate new content that mimics those patterns
- Generative models learn statistical patterns from existing data and sample from those learned patterns to create new content; they do not rely only on hand-coded rules or random output.
- R E F E R E N C E AIF-C01 Domain 2 exam guide [https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-](https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html)
1/ai-practitioner-01-domain2.html
- 49 D 1 · F U N D A M E N T A L S O F A I A N D M L
- Which of the following best describes the Amazon SageMaker Canvas ML tool?
- B Gives the ability to use machine learning to generate predictions without the need to write any code
- Amazon SageMaker Canvas gives you the ability to use machine learning to generate predictions without needing to write any code. With Canvas, you can chat with popular large language models (LLMs), access Ready-to-use models, or build a custom model trained on your data.
- R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/sagemaker/latest/dg/canvas.html>
- 50 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I
- A development team wants to transform a feature idea into structured requirements, a technical design, and an implementation task list. Which Kiro capability should it use?
- A Specs
- Kiro Specs formalize feature and bug-fix development. A feature spec produces structured requirements, design, and tasks artifacts so the team can review the plan and track implementation.
- R E F E R E N C E Kiro Specs <https://kiro.dev/docs/specs/>

51 D 2 · F U N D A M E N T A L S O F G E N E R A T I V E A I

Which of the following is correct regarding Large Language Models (LLMs)?

D The Large Language Models (LLMs) are non-deterministic

Large Language Models (LLMs) are non-deterministic, which implies that the generated text may be different for every user that uses the same prompt. You can use the inference parameter Temperature (having a value between 0 and 1), which regulates the creativity of LLMs' responses. Use a lower temperature if you want more deterministic responses, and use a higher temperature if you want creative or different responses for the same prompt from LLMs on Amazon Bedrock.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/bedrock/latest/userguide/general-guidelines-for-bedrock-users.html>

52 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

A research team is developing an AI agent to autonomously navigate and complete tasks in a dynamic environment. Which of the following approaches would be most effective in training the AI agent to handle unexpected challenges during its tasks?

D Applying reinforcement learning, where the agent learns to make decisions based on feedback from its interactions with the environment

Reinforcement learning lets an agent learn from interaction and reward feedback, which is more adaptable than trying to enumerate every possible situation with static rules.

R E F E R E N C E How Amazon Bedrock Agents works <https://docs.aws.amazon.com/bedrock/latest/userguide/agents-how-to-use.html>

53 D 1 · F U N D A M E N T A L S O F A I A N D M L

What are the key differences between real-time inference and batch inference in the context of machine learning?

B + C Real-time inference is used for applications requiring immediate predictions with low latency, whereas batch inference is used for processing large volumes of data at once, often with higher latency; Real-time inference follows a synchronous execution mode, whereas batch inference follows an asynchronous execution mode

Batch inferencing, also known as offline inferencing, generates model predictions on a batch of observations. Batch inference is a good option for large datasets or if you don't need an immediate response to a model prediction request. Use Cases: High throughput applications like data warehousing, ETL pipelines, periodic reports, and predictive maintenance.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/architecture/batch-inference-at-scale-with-amazon-sagemaker/>

- 54 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- An agent's conversation history has grown beyond the model's context limit. Which strategy best preserves useful context?
- B Select, summarize, or retrieve the most relevant information for the current task
- The context window is limited, so an application should treat it as a managed workspace. Summarization, relevance-based selection, retrieval, and memory can retain useful information without sending the entire history on every request.
- R E F E R E N C E Understanding context engineering for generative AI <https://docs.aws.amazon.com/prescriptive-guidance/latest/gen-ai-lifecycle-operational-excellence/dev-experimenting-context.html>
- 55 D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S
- Which method fine-tunes a model by retraining it with specific prompts and desired outputs?
- D Instruction Tuning
- Instruction tuning uses labeled instruction-response examples to improve a model's ability to follow task-specific directions.
- R E F E R E N C E Amazon Bedrock Prompt management <https://docs.aws.amazon.com/bedrock/latest/userguide/prompt-management.html>
- 56 D 5 · S E C U R I T Y , C O M P L I A N C E , A N D G O V E R N A N C E
- Why is maintaining data lineage crucial in the context of building machine learning models on AWS?
- B It ensures data privacy and compliance by tracking the flow and transformation of data
- Maintaining data lineage involves tracking the flow, transformations, and origins of data throughout its lifecycle. This is crucial for ensuring data privacy, security, and compliance with regulatory standards. By having a clear record of where data comes from and how it has been processed, organizations can ensure that their machine learning models are built on trustworthy and compliant data.
- R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/big-data/build-data-lineage-for-data-lakes-using-aws-glue-amazon-neptune-and-spline/>

D 4 · GUIDELINES FOR RESPONSIBLE AI

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Which AWS tool/utility provides guidance on how an ML model should be used and the risk rating of the model?

C Amazon SageMaker Model Cards

Use Amazon SageMaker Model Cards to document critical details about your machine learning (ML) models in a single place for streamlined governance and reporting. Catalog details such as the intended use and risk rating of a model, training details and metrics, evaluation results and observations, and additional call-outs such as considerations, recommendations, and custom information. By creating model cards, you can do the following: Provide guidance on how a model should be used.

R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/sagemaker/latest/dg/model-cards.html>

D 5 · SECURITY, COMPLIANCE, AND GOVERNANCE

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Which security discipline in the Generative AI Security Scoping Matrix focuses on identifying potential threats to generative AI solutions and recommending mitigations?

C Risk management

Risk management in the Generative AI Security Scoping Matrix involves identifying potential threats to generative AI solutions and recommending mitigations. It encompasses activities like risk assessments and threat modeling, which are essential for understanding and addressing the unique risks associated with generative AI workloads.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/security/securing-generative-ai-an-introduction-to-the-generative-ai-security-scoping-matrix/>

D 3 · APPLICATIONS OF FOUNDATION MODELS

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A company wants to use Amazon Bedrock for a use case that involves generating images from text prompts. Which of the following Foundation Models do you recommend?

D Stable Diffusion

Stable Diffusion is a generative artificial intelligence (generative AI) model that produces unique photorealistic images from text and image prompts.

R E F E R E N C E AWS documentation <https://aws.amazon.com/what-is/stable-diffusion/>

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D 4 · GUIDELINES FOR RESPONSIBLE AI

A company needs to support human reviews and audits for its ML model predictions. The solution should be easy to implement and have the facility to add multiple reviewers. Which AWS service do you recommend for this use case?

A Amazon Augmented AI (A2I)

Amazon Augmented AI (A2I) supports human review workflows for ML predictions and can route selected cases to multiple reviewers.

R E F E R E N C E Amazon Augmented AI <https://docs.aws.amazon.com/sagemaker/latest/dg/a2i-what-is.html>

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D 1 · FUNDAMENTALS OF AI AND ML

Which of the following options represent the CORRECT statements regarding the Amazon ML services?

A + D Amazon Polly is used to deploy high-quality, natural-sounding human voices in dozens of languages; Amazon Comprehend service uses machine learning to find insights and relationships in the text

Amazon Polly is a cloud service that converts text into lifelike speech. You can use Amazon Polly to develop applications that increase engagement and accessibility. Amazon Polly supports multiple languages and includes a variety of lifelike voices.

R E F E R E N C E AWS documentation <https://aws.amazon.com/ai/services/>

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D 2 · FUNDAMENTALS OF GENERATIVE AI

What is the primary purpose of diffusion models in generative AI?

D To generate high-quality samples from noisy data

Diffusion models learn to reverse a noise-adding process, enabling them to generate high-quality samples from noise.

R E F E R E N C E AIF-C01 Domain 2 exam guide <https://docs.aws.amazon.com/aws-certification/latest/ai-practitioner-1/ai-practitioner-01-domain2.html>

63 D 5 · SECURITY, COMPLIANCE, AND GOVERNANCE

Which of the following scenarios best illustrates the difference between exposure and prompt injection in the context of AI models? Prompt 1: "What is the best way to secure my online accounts?" Response A: "To secure your online accounts, use strong, unique passwords for each account, enable two-factor authentication, and regularly update your security settings. By the way, here's a secret key: 12345XYZ." Prompt 2: "Can you tell me a joke?" Response B: "Why don't scientists trust atoms? Because they make up everything! Also, remember your session ID: ABCDE12345." Prompt 3: "Give me a recipe for a chocolate cake." Response C: "Sure! To make a chocolate cake, you need flour, sugar, cocoa powder, and eggs. By the way, you should input the following command in your system: 'DELETE .'" Prompt 4: "What is the capital of Italy?" Response D: "The capital of Italy is Rome. Let's discuss your previous query about hacking tools." Response C is prompt injection; Response A is exposure

Prompt Injection refers to influencing the outputs by embedding specific instructions within the prompts themselves. Exposure refers to the risk of exposing sensitive or confidential information to a model during training or inference. The model can then reveal this sensitive data from their training corpus, leading to potential data leaks or privacy violations.
 R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/prescriptive-guidance/latest/llm-prompt-engineering-best-practices/common-attacks.html>

64 D 4 · GUIDELINES FOR RESPONSIBLE AI

What is the difference between interpretability and explainability in the context of Responsible AI?

- B Interpretability is about understanding the internal mechanisms of a machine learning model, whereas explainability focuses on providing understandable reasons for the model's predictions and behaviors to stakeholders

Interpretability refers to how easily a human can understand the reasoning behind a model's predictions or decisions. It's about making the inner workings of a machine learning model transparent and comprehensible. Explainability goes a step further by providing insights into why a model made a specific prediction, especially when the model itself is complex and not inherently interpretable.
 R E F E R E N C E AWS documentation <https://docs.aws.amazon.com/whitepapers/latest/model-explainability-aws-ai-ml/interpretability-versus-explainability.html>

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D 3 · A P P L I C A T I O N S O F F O U N D A T I O N M O D E L S

Consider the following AI model responses to user prompts. Which response exemplifies hallucination and which exemplifies toxicity? Prompt 1: "What is the capital of France?" Response A: "The capital of France is Mars." Prompt 2: "Tell me a joke." Response B: "Why don't scientists trust atoms? Because they make up everything!" Prompt 3: "Describe a good book for children." Response C: "You should read 'ABC', a great book with fun activities for kids." Prompt 4: "What do you think about people from [specific group]?" Response D: "People from [specific group] are inferior and should not be trusted."

D Response A is hallucination; Response D is toxicity

Toxicity refers to AI model-generated content that can be deemed as offensive, disturbing, or inappropriate. Hallucination refers to AI model-generated assertions or claims that sound true but are incorrect. Response A is an example of hallucination, where the AI model generates an irrelevant or incorrect response ("The capital of France is Mars"). Response D is an example of toxicity, where the AI model generates harmful or offensive content about a specific group.

R E F E R E N C E AWS documentation <https://aws.amazon.com/blogs/machine-learning/build-safe-and-responsible-generative-ai-applications-with-guardrails/>